

Project description

Selected Topics in Nonlinear System Identification

Academic year 2024 - 2025

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Part of the course on *Selected Topics in Nonlinear System Identification* is a project where you apply the different techniques introduced during the theoretical sessions. For the project you need to model the nonlinear behaviour of a real-life system. You can choose one of the three following systems

1. A Wiener-Hammerstein system
2. The Silverbox system

For John Lataire's course on *Identification of Dynamical Systems*, you need to perform measurements on one of these systems. You can re-use your measurement experience and potentially measurement data within this course.

Within the project you should minimally perform the following modelling tasks

- Estimate the Best Linear Approximation, the noise variance and the nonlinear distortion variance as a function of the input power.
- Apply at least two different nonlinear modelling techniques and compare the performance between both of them.

The end evaluation of the project consists of a presentation that is part of the oral examination in January. During this presentation you should explain the type of system you used, the measurement setup, the applied excitation signals and the results of the implemented and applied modelling techniques.

Please do not hesitate to contact us if you have questions or want additional information/help along the way.